Kaoru Sumi

Professor, Future University Hakodate

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Google Scholar kaoru-sumi-38862110

Research Profile

Researcher in affective computing and human-AI interaction, with research spanning persuasive technology, human-agent and human-robot interaction, embodied AI, VR/MR/XR, nonverbal communication, serious games, and affective learning. Current work focuses on socially embodied AI that can understand human affect and behavior, adapt its nonverbal and verbal behavior, and support learning, behavior change, communication, and social connection.

Research Areas

Areas: Affective Computing; Human-AI Interaction; Human-Agent Interaction; Human-Robot Interaction; Persuasive Technology; Embodied AI and Social Agents; VR/MR/XR; Serious Games and Affective Learning; Nonverbal and Multimodal Interaction; Physiological Sensing

Academic Appointments

Future University Hakodate, Professor, Department of Information Architecture — Hakodate, Japan, 2011 – present

Hitotsubashi University, Associate Professor — Tokyo, Japan, 2009 – 2011

National Institute of Information and Communications Technology (NICT), Researcher — Japan, 2004 – 2009

Osaka University, Assistant Professor — Osaka, Japan, 2000 – 2004

Communications Research Laboratory (CRL), Researcher — Japan, 1998 – 2000

ATR Media Information Science Laboratories, Special Research Fellow — Kyoto, Japan, 1996 – 1998

Visiting and International Positions

University of Alcalá, Invited Research Fellow, Giner de los Ríos Program — Spain, 2025
Conversational interaction in human-agent systems.

University of British Columbia, Visiting Professor — Vancouver, Canada, 2018

Education

Ph.D. **University of Tokyo**, Engineering — Japan, 1999
Dissertation: Research on Communication Support Using Individual Conceptual Spaces

M.S. **University of Tsukuba**, Management Systems Science — Japan, 1995

B.S. **Tokyo University of Science**, Physics — Japan 1988

Books and Edited Volumes

Affective Learning and Serious Games — 2026
Kaoru Sumi
10.1201/9781003782735 (CRC Press / Taylor & Francis)

Persuasive Technology: 21st International Conference, PERSUASIVE 2026, Hakodate, Japan, March 10-13, 2026, Proceedings — 2026
Kaoru Sumi, Raian Ali, Roberto Legaspi
10.1007/978-3-032-19687-3 (Springer, Lecture Notes in Computer Science, Vol. 16476)

Selected Publications

DXSP: Gesture-Only Temporal Localization of Decisive Moments in Dyadic Persuasion — 2026
Pakpoom Chaimook, Kaoru Sumi
10.3389/frai.2026.1928901 (Frontiers in Artificial Intelligence, 9, Article 1928901)

Emotional Expression in Persuasion by Quadruped Virtual Agents: Toward Cross-Species Design Patterns — 2026
Kaoru Sumi, Souki Osawa
10.48550/arXiv.2608.01895 (arXiv preprint, arXiv:2608.01895)

Designing Persuasive Interactions with Pet-Type Virtual Agents: Effects of Emotion and Context in Mixed Reality — 2025
Kaoru Sumi, Rio Harada
10.54941/ahfe1006912 (Human Factors in Design, Engineering, and Computing, AHFE 2025, Vol. 199, pp. 996-1005)

The Impact of Anonymity on Communication in the Metaverse — 2024
Yoshifuru Nakayama, Kaoru Sumi
10.1109/COMPSAC61105.2024.00335 (2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC), pp. 2095-2100)

Deception Detection in Educational AI: Challenges for Japanese Middle School Students in Interacting with Generative AI Robots — 2024
Ahmed Salem, Kaoru Sumi
10.3389/frai.2024.1493348 (Frontiers in Artificial Intelligence, 7, Article 1493348)

Persuasive Technology Through Behavior and Emotion with Pet-Type Artifacts — 2024
Rio Harada, Kaoru Sumi
10.1007/978-3-031-58226-4_12 (Persuasive Technology, PERSUASIVE 2024, LNCS 14636, pp. 151-160)

Experiences of Game-Based Learning and Reviewing History of the Experience Using Player's Emotions — 2022
Kaoru Sumi, Shusuke Sato
10.3389/frai.2022.874106 (Frontiers in Artificial Intelligence, 5, Article 874106)

Emotional Characteristic Analysis of Human Gait while Real-Time Movie Viewing — 2022
Nitchan Jianwattanapaisarn, Kaoru Sumi, Akira Utsumi, Nirattaya Khamsemanan, Cholwich Nattee
10.3389/frai.2022.989860 (Frontiers in Artificial Intelligence, 5, Article 989860)

Analysis and Prediction of Student Emotions while Doing Programming Exercises 2019
Thomas James Z. Tiam-Lee, Kaoru Sumi
10.1007/978-3-030-22244-4_4 (International Conference on Intelligent Tutoring Systems, Lecture Notes in Computer Science)

A Serious Game for Learning Social Networking Literacy by Flaming Experiences 2017
Kaoru Sumi, Kodai Kasai
10.1007/978-3-319-49616-0_3 (Intelligent Technologies for Interactive Entertainment, LNICST 178, pp. 23-33)

TelMe: A Personalized, Context-Aware Communication Support System 2001
Kaoru Sumi, Toyoaki Nishida
10.1109/5254.940022 (IEEE Intelligent Systems, 16(3), pp. 21-27)

Honors, Awards and Distinctions

- **IEEE Computer Society Distinguished Visitor (2026-2028).** Selected through a highly competitive process in recognition of contributions to IEEE, the computing profession, and the technical community. https://www.computer.org/profiles/kaoru-sumi
- **JSAI Research Meeting Outstanding Award**, Japanese Society for Artificial Intelligence, 2026, 2023, and 2017.
- **Best Presentation Award**, AIoT Behavior Transformation Research Meeting, 2025.
- **Best Technical Design Paper Award**, International Conference on Computers in Education (ICCE 2020).
- **Best Full Paper Award**, International Conference on Intelligent Tutoring Systems (ITS 2019).
- **IEEE Brain Winner and First Place Winner**, Brain-Computer Interface Hackathon at IEEE SMC 2018.

Selected Research Funding

NEDO / Strategic Innovation Promotion Program (SIP), Phase 3 2023-2025
Registered researcher, infrastructure technologies and governance for expanding the virtual economy.

JSPS KAKENHI, Grant-in-Aid for Scientific Research (C) 2018-2020
Principal Investigator - Content Creation and Sharing Platform for Simulation in Virtual Worlds.

JSPS KAKENHI, Challenging Exploratory Research 2015-2016
Principal Investigator - Visual Logic Programming for Training Logical Thinking.

JSPS KAKENHI, Challenging Exploratory Research 2012-2013
Principal Investigator - Service-Mind Computing through Facial-Expression and Behavioral Training.

JSPS KAKENHI, Grant-in-Aid for Scientific Research on Innovative Areas 2010-2011
Principal Investigator - Persuasive Technology through Human-Robot Interaction.

Keynote and Invited Talks

Keynote Speaker, IEEE TEMSMET 2026 Hyderabad, India
Keynote at IEEE TEMSMET 2026, October 29-31, 2026. Oct 2026

Invited Speaker, TED-HRI Workshop, IEEE RO-MAN 2026 Kitakyushu, Japan
Aug 2026

Keynote and Panel, IEEE NSS MIC RTSD 2025 2025
Designing Emotionally Persuasive and Inclusive AI: From Social Robots to XR Agents.

Keynote Speaker, Behavior Transformation by IoT International Workshop 2024 2024
Persuasive Technology and Emotions.

Plenary Speaker, Symposium on APEC ICT Skill Standards for AI 2023
Artificial Intelligence for the Future of Learning and Experiences in Training Students.

Teaching and Mentorship

- Designs and delivers undergraduate and graduate courses in interactive systems, AI, media, and human-agent interaction.
- Leads project-based learning in AI, VR/MR, robotics, games, and educational technology, and supervises doctoral, master's, and undergraduate research.
- Mentors domestic and international students through research-question formulation, system development, user studies, data analysis, conference presentation, and publication.

Editorial and Conference Leadership

- **Associate Editor**, Behaviour & Information Technology (Taylor & Francis), 2024-present.
- **Associate Editor**, Frontiers in Artificial Intelligence, AI for Human Learning and Behavior Change, 2025-present.
- **Guest Editor**, Frontiers Research Topic, Next-Generation Persuasive Technologies for Human-AI Interaction and Behavior Change, 2025-present.
- **Guest Editor**, JoVE Methods Collection, Interactive Technologies for Behavior Change and Emotional Engagement.
- **General Chair**, 21st International Conference on Persuasive Technology (PERSUASIVE 2026), Hakodate, Japan.

Professional and International Academic Service

- **IEEE Japan Council Women in Engineering Coordinator**, 2025-present.
- **Chair, IEEE Sapporo Section Women in Engineering**, 2020-present.
- **Executive Board Member, IEEE Sapporo Section**, 2023-present.
- **External Referee for Promotion and Tenure Review**, York University, Canada, 2026.
- Ph.D. dissertation committee service at De La Salle University, Philippines, and Thammasat University, Thailand.

Professional Memberships

- Senior Member, Institute of Electrical and Electronics Engineers (IEEE).
- Senior Member, Association for Computing Machinery (ACM).
- Member, Japanese Society for Artificial Intelligence (JSAI).
- Member, Information Processing Society of Japan (IPSJ).